BRIAN CAI

US Citizen | briancai@gatech.edu | 678-815-5089

Education

Georgia Institute of Technology
BS in Computer Science
Threads: Information Internetworks, Artificial Intelligence
May 2020 | GPA: 3.57

Skills

Proficient: Java, Python
Familiar: JavaScript, C, C#, C++, Typescript
Tools: Aurelia, Git, React, Gradle, Firebase, Spring Boot, Angular, Bash

Coursework

Core - OOP, Data Structures and Algorithms, Objects and Design, Algorithms Design and Analysis
Info/Networks - Computer Organization, Systems and Networks, Networking, Databases
Artificial Intelligence - Intro to AI, Robotics and Perception
Math - Discrete Math, Linear Algebra, Combinatorics & Graph Theory, Probability & Statistics
In Progress - Machine Learning for Trading, Automata and Complexity, Junior Design

Miscellaneous

Fencing
(Epee, USFA 'A' Rating)
Mandarin Fluent
Phi Kappa Theta
(Social Fraternity)
Dance Dance Revolution
(Stepmania)
CPR Certified
Summer Swim Coach (2016)
Policy Debate

Work Experience

The Home Depot | Software Engineer Intern
August 2018 - Present | Atlanta GA
- Conducted research and development at the "Orange Works" innovation center in Tech Square

Georgia Tech College of Computing | Teaching Assistant
August 2018 - Present | Atlanta GA
- CS 3630 Robotics and Perception - Assisted students during weekly office hours with implementing robotics software on Anki's Cozmo, including image recognition, localization, and pathing past obstacles to a goal (Python, Cozmo SDK, NumPy, SciPy)
- Graded students' lab demos and quizzes & provided feedback

The Home Depot | Software Engineer Intern
May 2018 - August 2018 | Atlanta GA
- Integrated custom payment system (Orange Pay) into new multibillion dollar order-building and checkout client (Order Up) that reduces training costs, checkout time, and fraud (Angular, Typescript, NPM, Yarn)
- Called enterprise RESTful API for prices and discounts (Java 8, Spring Boot, Gradle)
- Agile Methodology (Extreme Programming): paired programming, story writing, and extensive testing [TDD], including unit, integration, e2e, mutation, and performance (JUnit, Pitest, Gatling, Jasmine, Karma, Protractor)

MRI Software | Software Engineer Intern
May 2017 - August 2017 | Atlanta GA
- Modernized hundreds of web pages for real estate finance applications incorporating binding principles and responsive web-design for optimizing code (Aurelia, Javascript)
- Utilized Web APIs in refactoring code and altered SQL queries to populate the webpage forms in the front-end (C#, SQL, CSS, HTML)

HyTech Racing | Software Team Member
August 2016 – Present | Atlanta GA
- Designed and implemented electric racecar control systems software under an interdisciplinary team including CS, mechanical, EE, and business majors (Github: https://github.com/hytech-racing)
- Developed 4D systems dashboard (Workshop4 IDE) that reads CAN Bus communications from Arduinos (Teensy, C++, UX/UI)
- Developed Bluetooth and CAN libraries for Raspberry Pi and Arduino
- Formula Hybrid 2018 Electric Category Winner!

Projects

briancai.com | github.com/brian-cai

Shapeshifter Solver | April 2018
Neopets game solver with heuristic-guided graph search (A*). Game state processed with html parsers. Tools: Python, Beautiful Soup

Air/Mold Quality Database | Spring 2017
SQL database GUI using EER models and relational schemes. CS 4400 "Serve Learn Sustain" project. Tools: php, html, mySQLi

MealMe | HackATL 2016
Instagram inspired iOS App with food ordering
Tools: Heroku, Flask, Eatstreet/Yelp API, Ionic Framework (Typescript/Angular)

Can't Stump | VandyHacks 2016
AR Game starring Donald Trump as Slenderman. Won "Best Use" award and a free Project Tango. Tools: Google Project Tango, Unity, Audacity, Blender